

BURGAS, BulgariaWMO: **156550**Lat: **42.570N**Lon: **27.515E**Elev: **41**StdP: **100.83**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.4	-5.2	-11.9	1.4	-6.2	-9.2	1.7	-4.0	18.1	-1.1	15.1	-1.3	5.8	0	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	31.9	21.7	30.2	21.7	29.0	21.3	24.2	28.5	23.4	27.8	22.6	27.1	5.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	17.8	27.0	22.0	16.7	26.2	21.1	15.8	25.5	73.4	28.5	70.0	27.6	67.1	27.3	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.1	12.0	10.1	-10.4	35.5	2.2	2.5	-12.0	37.3	-13.3	38.8	-14.6	40.2	-16.2	42.0		
(5)			-11.0	26.0	2.2	0.9	-12.6	26.7	-13.9	27.2	-15.2	27.7	-16.8	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	2.8	4.4	7.5	11.5	16.6	21.3	23.8	23.8	19.5	14.2	9.4	4.3
	DBStd	8.24	4.38	4.57	4.05	3.47	3.11	2.68	2.07	2.30	2.95	3.63	4.36	4.50
	HDD10.0	767	227	163	100	23	1	0	0	0	0	8	62	184
	HDD18.3	2353	483	390	337	207	70	6	0	0	23	132	270	436
	CDD10.0	1969	2	6	22	67	207	339	428	427	284	139	43	7
	CDD18.3	514	0	0	0	1	17	95	170	169	56	6	1	0
	CDH23.3	3888	0	0	1	13	107	652	1382	1396	307	29	2	0
	CDH26.7	1065	0	0	0	1	17	158	417	417	52	3	0	0
Wind	WSAvg	4.2	4.1	4.5	4.6	4.3	4.0	3.9	4.1	4.4	4.4	4.3	4.0	4.2
Precipitation	PrecAvg	588	39	45	39	52	46	50	38	26	45	56	64	54
	PrecMax	763	93	113	90	96	111	133	111	86	161	200	187	113
	PrecMin	370	0	4	4	6	8	2	0	0	0	2	15	6
	PrecStd	101	28	27	22	20	26	34	30	21	39	44	37	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.8	18.9	22.1	25.2	28.8	32.2	34.1	34.1	30.5	26.2	23.0	17.9
		MCWB	11.7	13.2	13.9	16.2	19.6	22.1	21.3	22.2	19.8	18.5	16.4	13.7
	2%	DB	12.9	15.9	19.0	21.9	26.0	30.0	31.8	31.9	28.0	23.5	20.0	15.0
		MCWB	9.6	11.4	12.2	14.8	18.7	21.4	21.5	21.9	20.2	17.7	15.6	12.0
	5%	DB	10.2	13.2	16.3	19.1	24.1	28.2	30.1	30.2	26.2	21.5	17.4	12.5
		MCWB	8.0	9.7	11.1	13.6	17.8	20.8	21.6	22.0	19.7	17.0	14.5	10.5
	10%	DB	8.4	10.9	13.8	17.0	22.2	27.0	29.0	29.0	25.0	19.9	15.2	10.8
		MCWB	7.2	8.1	10.0	12.6	17.2	20.5	21.4	21.6	19.5	16.0	13.1	9.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	13.5	14.6	17.3	21.0	24.2	25.1	26.2	23.2	20.0	17.2	14.1
		MCDB	15.1	17.3	20.5	23.3	26.3	29.0	29.0	29.7	26.6	23.8	21.3	17.1
	2%	WB	10.1	11.8	12.8	15.5	19.8	22.9	24.1	24.6	21.9	18.6	15.9	12.5
		MCDB	12.4	15.3	17.9	20.6	24.4	27.6	28.5	28.7	25.5	22.0	18.9	14.7
	5%	WB	8.4	10.1	11.4	14.1	18.7	22.0	23.3	23.6	21.0	17.6	14.9	10.8
		MCDB	9.7	12.9	15.7	18.4	22.7	26.5	27.8	27.9	24.6	20.6	17.4	12.3
	10%	WB	7.2	8.4	10.2	13.1	17.7	21.2	22.5	22.8	20.1	16.6	13.5	9.4
		MCDB	8.2	10.5	13.3	16.3	21.4	25.7	27.1	27.2	24.0	19.3	15.3	10.7
Mean Daily Temperature Range	5% DB	MDBR	7.1	8.3	9.4	9.6	10.0	10.2	10.3	10.3	9.9	8.8	7.9	7.2
		MCDBR	9.8	12.2	13.9	12.9	12.4	12.1	11.8	11.8	11.6	11.4	10.4	9.2
		MCWBR	6.9	7.9	8.0	7.0	6.6	6.3	6.2	6.5	6.2	6.6	6.4	6.8
	5% WB	MCDBR	8.8	11.7	12.7	11.9	11.1	10.9	10.6	10.9	10.4	10.3	9.5	8.3
		MCWBR	6.6	7.9	7.6	6.8	6.2	6.1	6.0	6.3	6.0	6.3	5.8	6.5
Clear-Sky Solar Irradiance	taub	0.363	0.395	0.434	0.482	0.451	0.452	0.459	0.468	0.447	0.413	0.386	0.366	
	taud	2.298	2.204	2.121	2.000	2.172	2.216	2.212	2.178	2.200	2.263	2.275	2.286	
	Ebn at Noon	752	784	800	789	828	827	816	794	779	756	714	707	
	Edh at Noon	89	114	139	169	146	140	139	140	127	106	90	83	
All-Sky Solar Radiation	RadAvg	1.47	2.18	3.35	4.73	5.99	6.61	6.69	6.00	4.31	2.71	1.69	1.28	
	RadStd	0.19	0.26	0.32	0.48	0.40	0.49	0.40	0.29	0.38	0.29	0.22	0.19	